TFPT — Appendix H: the horizon unit system

One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code
(Hawking, de Sitter, Unruh, Page, scrambling — *standard physics in seam units, not new gravity*)

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What this document is about

A change of bookkeeping, not new gravitational physics: if gravity is the geometry-channel readout of the seam, then *all* horizons read the same boundary constant $c_3 = \frac{1}{8\pi}$. This note collects the readouts — Hawking, de Sitter and Unruh temperature, black-hole thermodynamics, the Page time, scrambling, the Nariai bound, $v_{\rm GW} = c$ and cosmic birefringence — in seam units, with two genuine compiler fingerprints ($1920 = |W(D_5)|$, $|\mu_4| = 4$).

The TFPT document set — what is treated where

Plain language: TFPT is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\rm car} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard Model, the constants and the scale grammar. The development is **five short documents plus this appendix**, best read in order:

1. `introduction` — reading guide, compiler closure, paper-by-paper comparison, predictions, the dependency DAG and proof ledger.
2. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction.
3. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.

H. `tfpt_horizon_readouts` — **Appendix H**: one seam constant as the universal horizon thermal code (a unit-system reframe, *not* a peer paper, *not* new physics).

You are reading Appendix H — a notation/unit reframing; all new-physics content lives in documents 1–4.

Contents

Scope and honest typing

If gravity is the geometry-channel readout of the seam (companion note "Seam response grammar"), then horizons are not separate miracles — they all read the *same* boundary normaliser $c_3 = 1/(8\pi)$. This note collects the readouts. Markers: [P] standard horizon physics rewritten in TFPT/seam units (no new claim), [I] an exact compiler connection (audit-level), [N] a numerical readout (reproduced), [A] a search target, not a result. Nothing here is sold as new gravitational physics; it is a *change of bookkeeping* that exposes shared structure.

Unit conventions used throughout

Three unit systems appear; the seam factor $c_3 = 1/(8\pi)$ is the same number in all of them, only the carried dimensionful constants change:

system	convention	Hawking temperature
Planck	$G{=}\hbar{=}c{=}k_B{=}1$	$T_H = c_3/M = 1/(8\pi M)$
reduced Planck	$\bar{M}_{\rm Pl} = M_{\rm Pl}/\sqrt{8\pi}$, $\hbar{=}c{=}k_B{=}1$	$T_H = c_3\,\bar{M}_{\rm Pl}^2/M$
SI	restore $G, \hbar, c, k_B$	$T_H = \dfrac{\hbar c^3}{8\pi G k_B M} = c_3\,\dfrac{\hbar c^3}{G k_B M}$

All boxed formulas below are in Planck units unless stated; the seam constant is dimensionless.

Standard references for the formulas reframed here: Hawking temperature and radiation (Hawking 1975); Bekenstein–Hawking entropy (Bekenstein 1973, Hawking 1975); Page time (Page 1993); fast scrambling $t_{\rm scr} \sim \beta \log S$ (Sekino–Susskind 2008, Maldacena–Shenker–Stanford 2016); Nariai bound (Nariai 1950); de Sitter thermodynamics (Gibbons–Hawking 1977). *Nothing below is a new derivation of these* — only a rewriting in the seam unit c_3.

1 The seam constant is the universal horizon temperature factor

Every Killing horizon has $T = \hbar\kappa/(2\pi c k_B)$. Since

$$\boxed{\frac{1}{2\pi} = 4c_3, \qquad \frac{1}{8\pi} = c_3} \qquad [\mathrm{I}],$$

the universal factor *is* the seam constant: $T_{\rm hor} = 4c_3\,\hbar\kappa/(ck_B)$. Black holes, de Sitter and Unruh therefore share *one* thermal grammar. [P] [`verification/v8_horizon.py`]

2 Schwarzschild thermodynamics in four c_3-lines

For a non-rotating, uncharged hole in Planck units ($G{=}\hbar{=}c{=}k_B{=}1$):

$$\boxed{T_H = \frac{c_3}{M}, \qquad S_{BH} = \frac{M^2}{2c_3}, \qquad P_H = \frac{c_3}{1920\,M^2}, \qquad \tau_{\rm evap} = \frac{640}{c_3}M^3} \qquad [\mathrm{P}].$$

Two of the constants are compiler numbers, not coincidences:

- the Hawking temperature factor is $c_3 = 1/(8\pi)$ — the same P1 boundary normaliser; [I]
- the radiated-power denominator is $1920 = |W(D_5)|$, the Weyl-group order of the D_5 carrier. [I]

Scope of P_H: this is the idealised massless single-channel blackbody normalisation. Greybody factors and the radiated species content shift the phenomenological evaporation coefficient; the c_3-form is the convention statement, not a general phenomenological prediction.

object	T_H	**lifetime / entropy**
solar-mass hole	6.17×10^{-8} K	$\tau_{\rm evap} \approx 2.1 \times 10^{67}$ yr; $S/k_B \approx 1.05 \times 10^{77}$
PBH evaporating now (1.7×10^{11} kg)	$k_B T \approx 61$ MeV	relevant for γ/ν PBH signatures

3 Page time and scrambling

Since $S_{BH} \propto M^2$ and $\tau \propto M^3$, the Page point ($M \to M/\sqrt{2}$) is

$$t_{\rm Page} = \Big(1 - \tfrac{1}{2\sqrt{2}}\Big)\tau_{\rm evap} \approx 0.6464\,\tau_{\rm evap} \qquad \text{[P]}.$$

In TFPT the Page curve is *boundary recoverability*: the physical sector is OS-reconstructed and gapped, and the recovered mutual information decays at the transfer rate, $I(A{:}C|B)_n \sim \lambda_2^n = (2/3)^{6n}$. **This is the same operator that builds the Standard Model:** the boundary transport spectrum is $\{1, (2/3)^6, (1/3)^6\}$ and its sub-leading eigenvalue $\lambda_2 = (2/3)^6$ is exactly the flavor-gap eigenvalue of Paper 2 ($\Delta_{\rm gap} = -\log(2/3)^6 = 6\log\frac{3}{2}$). One boundary transport thus governs *both* the SM flavor hierarchy and the horizon information recovery. [I] [`verification/v54_seam_horizon_keystones.py`] The scrambling prefactor carries a μ_4 signature: with $\beta = 1/T_H = 8\pi M$, $\beta/2\pi = 4M$, so

$$\boxed{t_{\rm scr} \sim |\mu_4|\, M \log S} \qquad (|\mu_4| = 4) \qquad \text{[I]-audit.}$$

4 De Sitter and the Nariai bound from the Λ closure

With the action-grammar Hubble/Λ readouts ($H_0/\bar{M}_{\rm Pl} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$, $\rho_\Lambda/\bar{M}_{\rm Pl}^4 \sim e^{-2\alpha_\star^{-1}}$), the asymptotic de Sitter horizon gives

$$\boxed{\frac{T_{dS}}{\bar{M}_{\rm Pl}} = 16c_3^2\, e^{-\alpha_\star^{-1}}, \qquad S_{dS} = \frac{e^{2\alpha_\star^{-1}}}{128c_3^4} = 32\pi^4 e^{2\alpha_\star^{-1}} \approx 3.32 \times 10^{122}} \qquad \text{[P]}.$$

This is the area-law horizon capacity of the universe from $(\alpha_\star^{-1}, c_3, H_\Lambda)$ alone. The maximal black hole in de Sitter (Nariai) is

$$M_N \approx 2.16 \times 10^{22}\, M_\odot \approx 4.3 \times 10^{52}\,\text{kg}, \qquad S_N = \tfrac{1}{3}S_{dS} \qquad \text{[P]},$$

the absolute gravitational-capacity limit, not an observed structure.

5 The maximal black hole is the anchor: SdS in seam units

The Nariai limit is not just a capacity number — written in compiler units, the entire Schwarzschild–de Sitter (SdS) family lands on atoms that are already load-bearing elsewhere in TFPT, with no adjustable parameter on either side. [I] [`verification/v101_horizon_anchor.py`]

Nariai = the anchor; the Koide 2/3 is the entropy bound [I] ([verification/v101_horizon_anchor.py])

With $f(r) = 1 - 2M/r - \Lambda r^2/3$, the horizon cubic at the maximal mass $M_N = 1/(3\sqrt{\Lambda}) = 1/(N_{\rm fam}\sqrt{\Lambda})$ is, in units $1/\sqrt{\Lambda}$,

$$\boxed{t^3 - 3t + 2 = (t-1)^2(t+2)} \qquad \text{roots } (1,1,-2),$$

with coefficients $(-3,2) = (-N_{\rm fam}, |\mathbb{Z}_2|)$ — and the root pattern is exactly the **traceless projection of the anchor**: $a - \frac{p_1(a)}{3}\mathbf{1} = -\frac{1}{3}(1,1,-2)$ for $a = (1,1,2)$ (compare $\chi_a = (t-1)^2(t-2)$, Paper 1). Five exact companions:

- **Entropy bound = the Koide branch value.** Each Nariai horizon carries $S_{dS}/3 = S_{dS}/N_{\rm fam}$ and the total is $\boxed{S_{\rm Nariai}/S_{dS} = 2/3 = |\mathbb{Z}_2|/N_{\rm fam}}$ (deficit exactly $S_{dS}/3$). The interpolation in $x = r_b/r_c$ is the closed form $S_{\rm tot}/S_{dS} = (x^2+1)/(x^2+x+1)$ — denominator $= \Phi_3(x)$, the $N_{\rm fam}$ cyclotomic — monotone from 1 to 2/3 with its minimum exactly at the merge (Fig. 2a).

- **Three-sheet conservation.** The cubic is traceless with fixed e_2, so $\pi\sum_i r_i^2 = 6\pi/\Lambda = |\mathbb{Z}_2|\, S_{dS}$ for *every* M: the black hole redistributes entropy between the two physical sheets and the virtual third root; the three-sheet total never moves.

- **Koide-form quotient of the roots.** $Q_{\rm geom} = \sum r_i^2/(\sum |r_i|)^2$ runs monotonically over $[\frac{3}{8}, \frac{1}{2}]$: pure dS gives $\frac{1}{2} = |\mathbb{Z}_2|/|\mu_4| = \delta$ (the half-step), Nariai gives $\frac{3}{8} = p_2(a)/e_1(a)^2$ — exactly the two nonzero $SU(4)_1$ conformal weights (Fig. 2b).

- **The mass line is a double cover.** $\mathrm{disc}_r \propto (1-3m)(1+3m)$ in $m = M\sqrt{\Lambda}$: branch points $m = \pm 1/N_{\rm fam}$, separation $2/3 = |\mathbb{Z}_2|/N_{\rm fam}$, deck involution = the black-hole ↔ cosmological horizon swap — the gravitational twin of the flavor sheet $\mathbb{Z}_2$ and of the split $N_{\rm fam}$-linear cover $(3x+2)(3x+5)$ of Paper 2 (Fig. 1).

- **Temperature lemma / orientation.** $|\kappa_b/\kappa_c| = (2x+1)/(x(x+2)) > 1$ for $x < 1$ (equality iff $x = 1$): the black-hole sheet is *always* hotter, so evaporation flows *away* from the merge — the anchor configuration is the **repeller** and the democratic endpoint the attractor, the same orientation as the flavor flow (Koide attractor / carrier repeller, Paper 4). *Audit reading.*

Completing the seam-unit table: $dM = c_3\,\kappa\, dA$ (first law), $M = 2c_3\,\kappa A$ (Smarr), $S \le ER/(4c_3)$ (Bekenstein), $P_H = c_3/(1920\, M^2)$, lifetime $\tau = 5120\pi M^3 = 128\, g_{\rm car} M^3/c_3$ ($128 = 2^7$ the dS prefactor, 7 the scalaron exponent; photon/geometric-optics conventions, typed external), Kerr $A_{\rm ext} = M^2/c_3$ with $A_{\rm ext}/A_{\rm Schw} = 1/2 = 1/|\mathbb{Z}_2|$ and $S_{\rm ext} = M^2/(4c_3)$, and the Bekenstein–Mukhanov–Hod area quantum $4\ln 3 = \ln 81 = \ln(N_{\rm fam}^4) = \ln(\text{disc of the flavor cover})$ ([P] as the QNM reading). *Honest scope:* the GR identities are textbook facts; the [I] content is six independent landings on already-load-bearing atoms with zero free parameters. The "carrier in the bulk" reading (two glue chiralities = two horizon sheets, ramification = Nariai) stays [P]. No new data test (M_N today $\sim 10^{22} M_\odot$): a structure surface, not a measurement surface.

One orientation: the anchor is the *stationary repeller* in both sectors [I]/[P] ([verification/v102_seam_orientation.py])

The evaporation direction of the temperature lemma upgrades to exact variational structure on both one-dimensional moduli lines (Fig. 4). **Flavor:** the canonical relaxation $\dot{q} =$

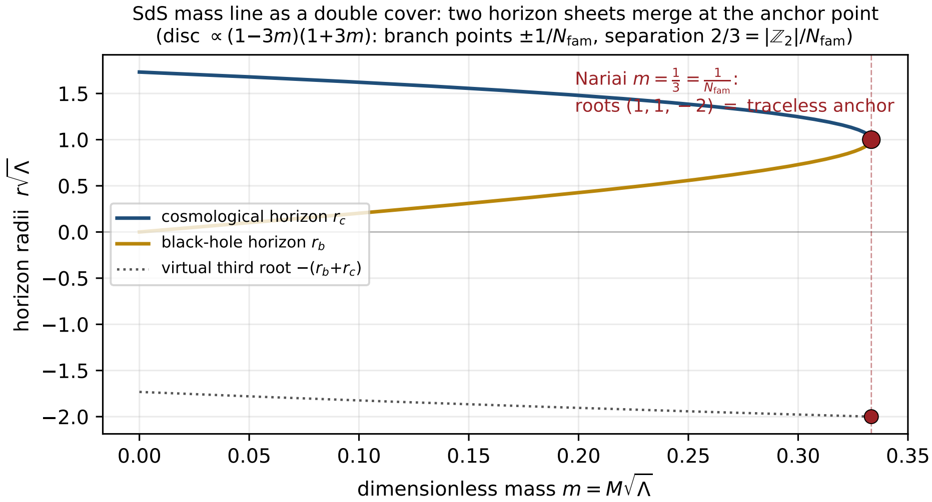

Figure 1: The SdS mass line as a double cover: black-hole and cosmological horizon sheets merge at the Nariai point $m = 1/N_{\text{fam}}$, where the three roots form the traceless anchor pattern $(1, 1, -2)$. [`verification/v101_horizon_anchor.py`]

$(\Delta/N_{\text{fam}})(q-2)(q-5)$ is the *gradient flow* of the cubic potential

$$V(q) = -\frac{\Delta}{N_{\text{fam}}}\Big(\frac{q^3}{3} - \frac{7}{2}q^2 + 10q\Big), \qquad \boxed{V''(2) = +\Delta, \quad V''(5) = -\Delta}$$

— the critical points are exactly the two branch points, the stationary curvatures are ±the transfer gap, the inflection sits at $q = \frac{7}{2}$ = scalaron/2, and the Lyapunov rate is constant: $d(-\ln\rho)/dt = \Delta$ exactly. **Gravity:** the SdS entropy functional has

$$\frac{d}{dx}\frac{S_{\text{tot}}}{S_{dS}} = \frac{(x-1)(x+1)}{\Phi_3(x)^2}, \qquad \boxed{\Big(\frac{S_{\text{tot}}}{S_{dS}}\Big)''(1) = \frac{2}{9} = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}^2} = \frac{2}{3}\cdot\frac{1}{3}}$$

— the Nariai/anchor point is the *unique* physical stationary point, its curvature is a grammar constant (= the product of the two Nariai entropy fractions), and the temperature lemma makes the evaporation an entropy *ascent* away from it. **Orientation theorem [I]:** in both sectors the anchor-type configuration is a stationary point of the natural functional and the dynamics flows away from it, with grammar-constant stationary curvatures. *Honest disanalogies (recorded):* the flavor attractor is itself a branch point while the gravity attractor is the smooth endpoint; the flavor functional is the potential of a [P]-typed relaxation while the gravity functional is the Bekenstein–Hawking entropy — "one variational principle of the seam" stays a *reading* [P], and no open gate is touched.

The trisection normal form: the canonical coordinate exists [I] ([`verification/v103_trisection_normal_form.py`])

The curvature-matching question of the previous box has a closed answer. The SdS horizon cubic is uniformized by **angle trisection**: with $r = 2\cos\theta$ (units $1/\sqrt{\Lambda}$), $r^3 - 3r = 2\cos 3\theta$ *exactly*, so the cubic is $\cos 3\theta = -3m$ — the three horizon roots are the three trisection branches (the $\mathbb{Z}_3$ rotation = the triality deck, cf. $\operatorname{coker} Q = \mathbb{Z}/N_{\text{fam}}$). In the centered angle

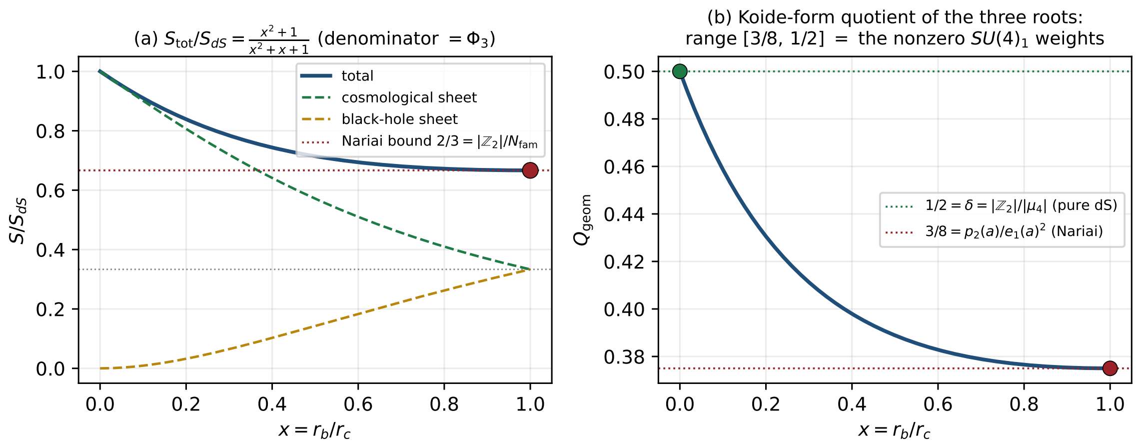

Figure 2: (a) Total SdS entropy interpolates from S_{dS} to the Nariai bound $\frac{2}{3}S_{dS}$ via $(x^2{+}1)/\Phi_3(x)$. (b) The Koide-form quotient of the three roots runs exactly over the nonzero $SU(4)_1$ weights $[\frac{3}{8}, \frac{1}{2}]$. [`verification/v101_horizon_anchor.py`]

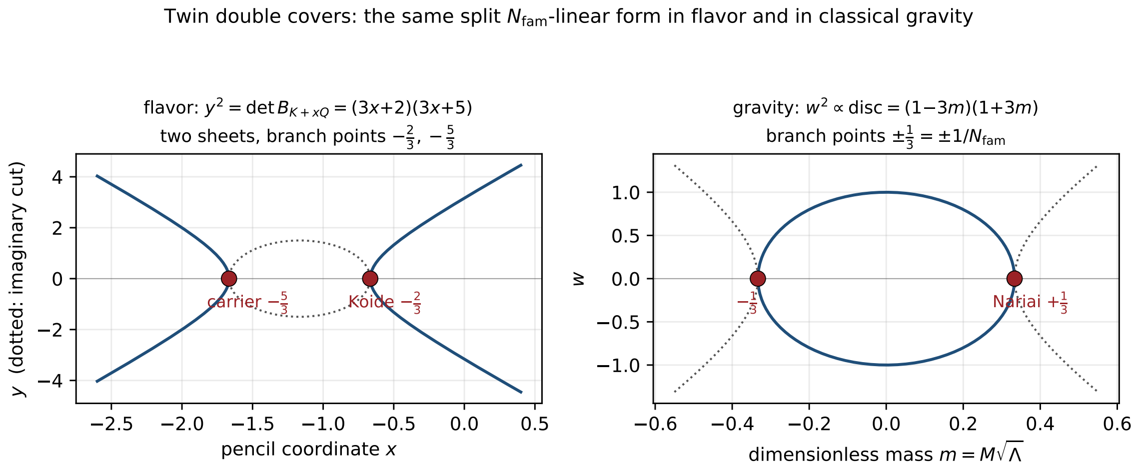

Figure 3: Twin double covers: the flavor anchor-block cover $y^2 = (3x{+}2)(3x{+}5)$ (Paper 2) and the SdS mass-line cover $w^2 \propto (1{-}3m)(1{+}3m)$ share the split N_{fam}-linear form with rational branch points at spine-over-family values. [`verification/v101_horizon_anchor.py`]

ψ ($\cos\varphi = -3m$, $\psi = \varphi - \pi$; Nariai $\psi = 0$, pure dS $\psi = \pi/2$, deck $\psi \to -\psi$):

$$m = \frac{\cos\psi}{N_{\rm fam}}, \qquad \boxed{\frac{S_{\rm tot}}{S_{dS}} = \frac{4}{3} - \frac{2}{3}\cos\frac{2\psi}{3}}$$

— *one* cosine whose mean ($|\mu_4|/N_{\rm fam}$), amplitude and frequency (both $|\mathbb{Z}_2|/N_{\rm fam}$) are glue atoms over the family count (Fig. 5). Three exact consequences:

- **Canonical curvature = the Koide constant to the family power:** $\sigma''(\psi{=}0) = \frac{8}{27} = (\frac{2}{3})^3 = (|\mathbb{Z}_2|/N_{\rm fam})^{N_{\rm fam}}$.
- **Invariant base slope:** $d\sigma/dm|_{\rm Nariai} = -\frac{8}{9} = -\operatorname{rank} E_8/N_{\rm fam}^2$ (coordinate-invariant, cross-checked in both coordinates; generally $-\frac{4}{3}\sin(2\psi/3)/\sin\psi$); dimensionful $dS_{\rm tot}/dM|_N = -r_N/(N_{\rm fam}c_3)$.
- **The bridge:** the flavor invariant is a *rate*, multiplier $(2/3)^6 = (2/3)^{2N_{\rm fam}}$ per transport step; the gravity invariant is a *curvature*, $(2/3)^{N_{\rm fam}}$ in the canonical angle — same base

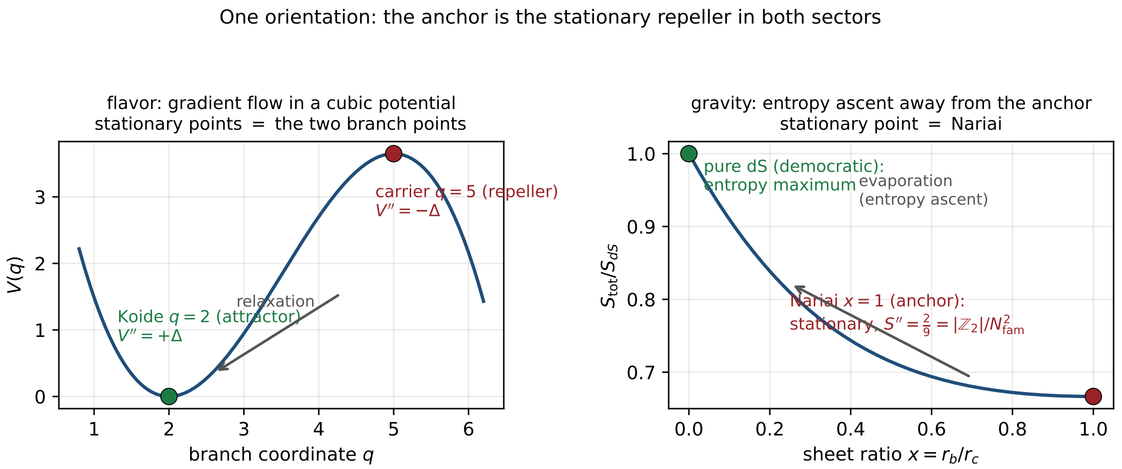

Figure 4: The orientation theorem: the anchor configuration is the stationary repeller of the natural functional in both sectors — flavor (cubic potential, curvatures $\pm\Delta$) and gravity (SdS entropy, curvature $2/9=|\mathbb{Z}_2|/N^2_{\mathrm{fam}}$). [`verification/v102_seam_orientation.py`]

$2/3=|\mathbb{Z}_2|/N_{\mathrm{fam}}$, exponent ratio $2=|\mathbb{Z}_2|$. What is still missing for a full identification is the gravity-side *clock* (the near-Nariai evaporation generator that converts curvature into a rate) — the same class of missing object as the Koide transfer generator of Paper 4: **one clock, two known geometries**. That identification stays [P].

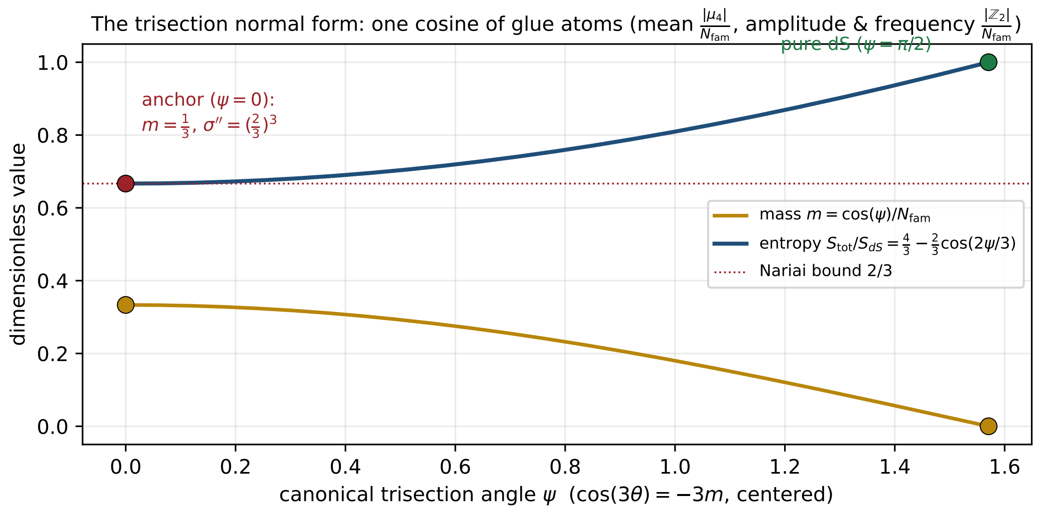

Figure 5: The trisection normal form: in the canonical angle ψ the mass and the total entropy are pure (co)sines built from glue atoms; the anchor sits at $\psi=0$ with canonical curvature $(2/3)^3$. [`verification/v103_trisection_normal_form.py`]

The classical clock speaks anchor — and the honest (2/3)-test [I]/[P] ([`verification/v104_nariai_clock.py`])

The clock question of the previous box has a *classical* half, and it is pure GR (Ginsparg–Perry 1983; Bousso–Hawking class): linearize around the Nariai geometry $dS_2\times S^2$. **Static**

pin [I]**:** on the dS_2 static patch the function $\phi(\rho) = \rho$ satisfies $\frac{d}{d\rho}[(1 - \Lambda\rho^2)\phi'] = -2\Lambda\phi$ *exactly* — and this static mode *is* the exact SdS two-horizon deformation, so the S^2-radius modulus mass is pinned internally: $m^2 = -2\Lambda = -|\mathbb{Z}_2|\Lambda$ (the Laplace-type form of the linearization is the one external GR import, typed honestly). The Euclidean spectrum is the Ginsparg–Perry tower $(l(l+1) - 2)\Lambda$: exactly *one* negative mode $-|\mathbb{Z}_2|\Lambda$. **The clock** [I]**:** in Hubble units the homogeneous modes $\phi \sim e^{\lambda H t}$ obey

$$\boxed{\chi_{\text{clock}}(\lambda) = \lambda^2 + \lambda - 2 = (\lambda - 1)(\lambda + 2)}$$

— **the anchor quadratic**: eigenvalues $\{1, -2\}$ = the distinct anchor roots, and the Nariai cubic factors as $(t-1)^2(t+2) = (t-1) \cdot \chi_{\text{clock}}(t)$. The anchor thus appears a *third* time: as configuration roots, as the curvature base, and as the spectrum of the linearized clock. The horizon split is linear in the canonical angle with coefficient $1/\sqrt{N_{\text{fam}}}$, and the entropy deviation grows at exactly $2H = |\mathbb{Z}_2|\times$Hubble. **Honest** (2/3)**-test** [P]**:** the classical clock eigenvalues are *integers*, not (2/3)-powers — the (2/3)-powers stay in the functional (curvature $(2/3)^{N_{\text{fam}}}$). The conversion curvature→rate at one loop (quantum (anti-)evaporation) remains external QFT input: the bridge to the flavor rate $(2/3)^{2N_{\text{fam}}}$ is *not* closed here. The seam variational principle now has exactly *one* missing object: the quantum clock.

The quantum-clock target, made quantitative [I]/[P] ([verification/v107_quantum_clock_target.py])

Instead of guessing the quantum clock, its job description can be pinned exactly. **(1) The classical clock already lives on the cusp ladder** [I]**:** across the physical Ginsparg–Perry modes ($l = 0$: $(\lambda-1)(\lambda+2)$; $l = 1$: $\lambda(\lambda+1)$) the decay exponents are

$$\boxed{\{0, -1, -2\} = -\operatorname{Spec}(V) = -N_{\text{fam}} \times \text{cusp weights}}$$

— exactly the grading that carries the transfer operator's eigenspaces (Papers 2/6); the $l \geq 2$ modes are damped oscillations with $\operatorname{Re}\lambda = -\frac{1}{2} = -\delta$ (audit). **(2) The exact target** [I]**:** the transfer rates $\{0, \Delta, 6\ln 3\}$ are *not* weight-linear — rate(2)/rate(1) $= \log_{3/2} 3 = 1 + \log_{3/2} 2 = 2.7095$, with the exact identity $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$: *the quantum clock must bend the weight-linear spectrum by* $\log_{9/4} 3 = 1.35476$ *per weight step.* **(3) Coupling landscape** [I]**:** the dimensionless clock coupling $\kappa = (c/24\pi)(\Lambda/\bar{M}_{\text{Pl}}^2)$ with $c = 8$ is 7.6×10^{-122} at the cosmological Nariai (hopeless) and $\kappa = 1/(3\pi) = 0.106$ at seam curvature — the only regime where an $O(0.35)$ bend is reachable, and it is borderline non-perturbative: one-loop linear response cannot fix the bend; an $O(1)$ resummation or an exact seam construction is required. *Status [P]:* R1 is sharpened from "find a clock" to "produce the bend $\log_{3/2} 3$ on the cusp ladder at coupling $O(1/3\pi)$". Nothing is fitted; the identification stays open.

6 Turnaround radius, gravitational waves, light

Bound structures. In a Λ-universe the maximal bound radius is $r_{\text{ta}}(M) = (GM[2\pi e^{\alpha_\star^{-1}}/\bar{M}_{\text{Pl}}]^2)^{1/3}$ — halo, cluster, supercluster and void scales follow from the Λ closure with no new parameter. [P]

Gravitational waves. The low-curvature branch is $R + R^2$; the tensor graviton runs on the Lorentz cone and the only extra mode (the scalaron, $M_{\text{scal}} \approx 3.06 \times 10^{13}$ GeV) is frozen for astrophysical frequencies. Hence

$$\boxed{v_{\text{GW}} = c} \qquad \text{(no measurable dispersion)} \qquad \text{[P]},$$

Black-hole mechanics in seam units ($c_3 = 1/(8\pi)$): every coefficient is a compiler atom

quantity	standard form	seam form	compiler atom
Einstein equations	$G_{\mu\nu} = 8\pi T_{\mu\nu}$	$G_{\mu\nu} = T_{\mu\nu}/c_3$	$c_3 = 1/(8\pi)$ (P1)
first law	$dM = \frac{\kappa}{8\pi} dA$	$dM = c_3\,\kappa\,dA$	seam constant
Smarr (Schw.)	$M = \frac{\kappa A}{4\pi}$	$M = 2c_3\,\kappa A$	
temperature	$T_H = \frac{\kappa}{2\pi}$	$T_H = 4c_3\,\kappa$	$1/(2\pi) = 4c_3$
entropy	$S = \frac{A}{4}$	$S = 2\pi c_3 A$	$1/4 = 1/\lvert\mu_4\rvert$
Bekenstein bound	$S \le 2\pi ER$	$S \le ER/(4c_3)$	
Hawking power	$P = \frac{1}{15360\,\pi M^2}$	$P = \frac{c_3}{1920\,M^2}$	$1920 = \lvert W(D_5)\rvert$
lifetime	$\tau = 5120\,\pi M^3$	$\tau = 128\,g_{\mathrm{car}} M^3/c_3$	$128 = 2^7$, $7 =$ scalaron
Kerr extremal	$A_{\mathrm{ext}} = 8\pi M^2$	$A_{\mathrm{ext}} = M^2/c_3$	$A_{\mathrm{ext}}/A_{\mathrm{Schw}} = 1/\lvert\mathbb{Z}_2\rvert$
area quantum (Hod)	$\Delta A = 4\ln 3$	$\Delta A = \ln(N_{\mathrm{fam}}^4)$	$81 =$ disc of the cover
Nariai bound	$S_N = \frac{2\pi}{\Lambda}$	$S_N = \frac{2}{3} S_{dS}$	$2/3 = \lvert\mathbb{Z}_2\rvert/N_{\mathrm{fam}}$ (Koide)

Figure 6: Black-hole mechanics in seam units: every classical coefficient is a compiler atom. [`verification/v101_horizon_anchor.py`]

a clean falsifier: a genuine $v_{\mathrm{GW}} \neq c$ would break this gravity closure.

Light. TFPT does not predict the SI number c (definitional); it predicts the *universality of the causal cone* — RP → OS reconstruction → one Lorentzian cone shared by EM, gravity and massless modes ($v_\gamma = v_g = c$, no native photon dispersion). What it *does* compute is not a speed shift but a polarisation rotation, cosmic birefringence

$$\boxed{\beta_{\mathrm{rad}} = \frac{u}{4\pi} \approx 0.2424^\circ} \qquad \text{[N]},$$

from the same seed $u = \phi_0^{\mathrm{ret}}$ that fixes the Cabibbo angle, Ω_b and θ_{13}.

7 Search targets (not claims)

Audit-level search ansätze [A]

- **Black-hole echoes:** any near-horizon echo amplitude ratio would be a natural transfer-rate target, $\mathcal{A}_{n+1}/\mathcal{A}_n \lesssim (2/3)^6 \approx 0.0878$. A search ansatz, no prediction.
- **Quantum recovery:** the Page-curve recovery kernel $I \sim (2/3)^{6n}$ is a falsifiable shape if a boundary-recovery rate is ever measured.

Net

All horizon formulas share the one seam normalisation $1/(2\pi) = 4c_3$. Black-hole, de Sitter and Unruh temperatures, the Page time, scrambling, the Nariai bound, turnaround radii, $v_{\mathrm{GW}} = c$ and birefringence are *readouts of one seam constant* — standard physics rewritten in TFPT units, with two genuine compiler fingerprints ($1920 = |W(D_5)|$ in Hawking power,

$|\mu_4| = 4$ in scrambling). No new gravitational physics is claimed; the value of the reframe is that horizons stop being separate modules.

Appendix: computational verification

The seam-unit identities of this note (the thermal grammar $\frac{1}{2\pi}$=$4c_3$, the power denominator 1920=$|W(D_5)|$, the de Sitter entropy form, the Page fraction, the scrambling $|\mu_4|$=4, and $\beta_{\rm rad}$=$\varphi_0^{\rm ret}/4\pi$=0.2424°) are re-derived and machine-checked in `verification/v8_horizon.py`. Run `python verification/v8_horizon.py` (needs `mpmath`); it imports the same c_3=1/(8π) primitive as the rest of the suite. The search-target ansätze remain [A] and are deliberately not part of the pass/fail checks.